Chapter II Review of Related Literature and Studies

A. Semantic Segmentation in Agricultural Diagnostics

Traditional agricultural disease monitoring encompasses a wide range of established diagnostic methods, from visual observation and microscopy to immunological and molecular genetic identification, all of which remain significant in phytopathological research for enabling timely and accurate disease control decisions (Khakimov et al., 2022). However, despite the wide range of available techniques, field-level practice continues to rely predominantly on manual inspection by farmers and agronomists. While experienced practitioners are able to identify diseases accurately, this continued reliance on unaided manual inspection is considered to be time-consuming, labor-intensive, and highly dependent on expertise, and visual disease assessment is itself prone to substantial inter- and intra-rater variability and subjective interpretation (Bock et al., 2022; Koshariya et al., 2025). These limitations, when placed at a larger scale across thousands of hectares and diverse crop cultivars, establish a clear demand for a scalable, objective, and automated diagnostic method capable of operating consistently under real-world field conditions.

The early stages of establishing an automated diagnostic method of detecting plant diseases relied on traditional digital image processing and classical machine learning algorithms. Before the utilization of deep learning, the standard pipeline for machine learning based techniques in plant disease detection involved heavy usage of feature engineering in order to extract relevant information from raw images (Hassan et al., 2022).

Table 2.1 Pipeline for identifying plant diseases (Hassan et al., 2022)

Early automated approaches relied primarily on color-based feature extraction for plant disease detection which helped determine disease via visually distinct chromatic changes (e.g: yellowing, browning, dark spots) that are distinguishable from healthy green tissue (Hassan et al., 2022). Color space transformations such as YCbCr, HSI, and CIELAB were commonly utilized to amplify these chromatic differences helping make diseased regions more distinguishable from healthy tissue (Hassan et al., 2022). These methods were combined with classical segmentation techniques such as Otsu's thresholding (Otsu, 1979), an approach that early studies applied to isolate diseased spots from leaf images (Chaudhary et al., 2012).

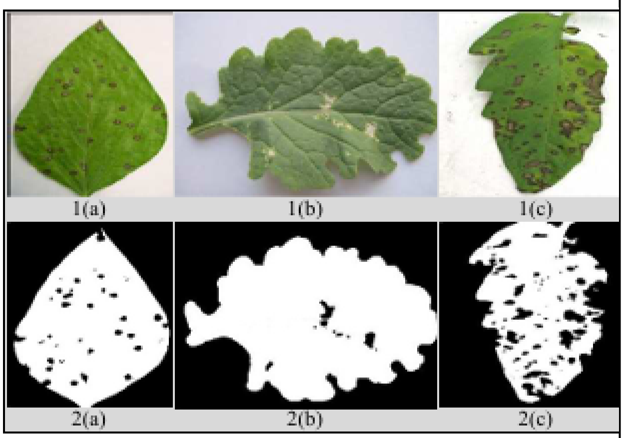

Figure 2.1 Leaf Spot Lesion Results (Chaudhary et al., 2012)

Other methods such as K-means clustering were also explored, as demonstrated by Pugoy & Mariano (2011), who applied the method to rice leaf disease detection across multiple disease types. However, these color-dependent approaches were limited in scope as it can only effectively detect diseases manifested through visible chromatic change and is unable to capture symptoms that are defined by properties such as lesion shape, surface texture, or spatial distribution (Hassan et al., 2022). Which necessitates the incorporation of shape and texture based disease detection for subsequent approaches (Hassan et al., 2022).

Recognizing that color-based approaches were insufficient for diseases that manifested through shape or texture, future methodologies involved the incorporation of shape and texture descriptors which allowed researchers to capture patterns that chromatic features cannot represent (Hassan et al., 2022; Islam et al., 2017). Texture descriptors derived from the Gray Level Co-occurrence Matrix (GLCM), originally introduced by Haralick et al. (1973), capture surface properties such as roughness, contrast, and uniformity that chromatic features cannot represent. In a representative classical pipeline, Islam et al. (2017) combined image segmentation with a multiclass support vector machine to classify potato diseases, reporting approximately 95% accuracy. However, determining the appropriate threshold values for their method was noted to be a challenge as each stage of the pipeline from segmentation to feature extraction to classification, relied on parameters that were manually defined rather than learned from data making the pipeline inherently brittle (Islam et al., 2017; Hassan et al., 2022). A configuration created for one crop, disease type, or imaging condition rarely can rarely be transferred reliably to another which means that generalizability across diverse real-world agricultural settings remains a persistent and unresolved challenge (Hassan et al., 2022).

Mohanty et al., (2016) established an early benchmark for the application of convolutional neural networks in image-based plant disease detection, demonstrating that deep learning models trained on large-scale datasets could achieve strong diagnostic performance. Early implementations in this domain primarily focused on Image Classification, as highlighted by Sharma et al., (2021), traditional image classification approaches served as the early standard for automating diagnostics, wherein the convolutional neural network learns hidden patterns across the entire image to predict a singular disease class eventually producing an output with a single categorical diagnosis. The paper itself utilized image classification in order to identify rice and potato plant leaf diseases leading to a 99.58% accuracy with classifying rice images and a 97.66% accuracy with potato leaves (Sharma et al., 2021). While studies that utilize classification-based networks have demonstrated considerable efficacy in binary diagnostic tasks, they are noted to possess a critical spatial limitation (Das et al., 2024; Nyawose et al., 2025). Classification models provide no spatial context regarding the location of an infection or the extent of its spread across plant tissue (Wang et al., 2025; Das et al., 2024). This limitation is consequential for the present study, as accurately estimating disease severity is more reliably achieved through the isolation of specific lesion boundaries which image-level classification cannot support, and one that necessitates a transition toward more spatially precise computer vision techniques (Gonçalves et al., 2021).

To address the spatial limitations of image-level classification, recent studies have begun to utilize semantic segmentation as a more precise diagnostic framework. Semantic segmentation, as described by Ren et al., (2025) as the process of assigning a predefined semantic label to every pixel within an image, allows models to describe precise boundaries between healthy tissue, background regions, and symptomatic lesion areas, helping provide a level of spatial resolution that classification-bound approaches cannot offer (Gonçalves et al., 2021; Wei et al., 2026). Early work by Barbedo (2017) demonstrates the viability of automated symptom segmentation in digital leaf photographs, showcasing that pixel-level descriptions of diseased regions could support the quantitative measurement of infected surface areas, which is an important metric in determining disease severity estimation. Subsequent deep learning-based approaches have expanded on this foundation, with Shoaib et al., (2022) demonstrating that lesion segmentation models could simultaneously identify lesion boundaries, distinguish between disease subtypes and support survival probability estimation. More recently, Yang et al., (2024) expanded segmentation to real-world field environments, addressing challenges introduced by complex backgrounds, variable lighting, and multi-scale disease presentation. The present study builds on these established studies, utilizing semantic segmentation to facilitate accurate lesion boundary descriptions on complex, in-the-wild datasets (Barbedo, 2017; Yang et al., 2024).

State-of-the-art segmentation architectures such as SegNeXt, proposed by Guo et al., (2022), leverage advanced convolutional attention mechanisms to achieve strong segmentation performance across complex scene understanding tasks. Although SegNeXt has better inference time performance compared to prior models that are purely attention-based segmentation networks, it remains a memory and parameter heavy architecture and is unsuitable to resource-constrained environments without compression (Guo et al., 2022; Yu et al., 2025). Yu et al. (2025) observe that the computational power and electricity supplied by deployed IoT agricultural devices are limited, making computational efficiency crucial for practical deployments. This bottleneck between high-capacity segmentation models' computational needs and the constraints on practical edge devices (Yu et al., 2025) motivates the current study, which examines how well a lightweight student model performs segmentation after knowledge distillation and INT8 quantization on the PlantSeg dataset.

B. SegNeXt

Recent developments in computer vision have been marked by the dominance of transformer architectures in pixel-level image categorization. However, this trend has been challenged by the introduction of SegNeXt, which is based on an optimized convolutional framework that delivers comparable efficiency and contextual data processing to state-of-the-art models (Guo et al., 2022). The architecture is characterized by the ability to extract complex, multi-scale information without relying on computationally expensive self-attention mechanisms. To address the cost of standard self-attention, the MSCA module is designed. Multi-branch depthwise strip convolutions are integrated with channel mixing. This technique ensures that relevant spatially sensitive feature elements are selectively prioritized by MSCA while minimizing computation. As it processes higher encoder stages during the decoding phase, the Hamburger module efficiently extracts features. This is because Hamburger has a simple design and effectively captures long-range spatial context while avoiding the computational costs of more complex attention-based decoder mechanisms (Guo et al., 2022). On benchmark datasets, SegNeXt yields strong results relative to some models that possess larger numbers of parameters. Specifically , on Pascal VOC 2012 test, the largest SegNeXt variant reaches an mIoU of 90.6% while using dramatically less computation than NAS-FPN EfficientNet-L2 (Guo et al., 2022). Guo et al. (2022) suggest this further points toward an efficient segmentation design.

The network achieves a 2.0% improvement in average mIoU compared to the prior state-of-the-art methods on ADE20K, while also maintaining or reducing FLOPs/MACs on the same dataset (Guo et al., 2022). When compared on Cityscapes dataset where urban areas contain detailed textures and objects, the smaller SegNeXt-S variant outperforms the SegFormer-B2 model, with about half the parameters and lower FLOPs/MACs (Guo et al., 2022). In Guo et al. (2022), the results show that SegNeXt offers higher mIoU while being more computationally efficient than other relevant comparable transformer-based models. This shows a good trade-off between high capacity and computational efficiency, making SegNeXt a feasible high-capacity teacher in a distillation setup. In this study, the SegNeXt-B configuration with the MSCAN-B backbone is adopted as this high-capacity teacher (Guo et al., 2022). Nevertheless, SegNeXt's capabilities in terms of real-time or edge deployment are not evaluated in this study. The only use case is a high-capacity teacher model from which knowledge is transferred to the lighter student model. Even though it showed strong performance on standard segmentation benchmarks, it has limited effectiveness when there are problems with visual qualities of the original images (Wei et al., 2026). Specific failure scenarios documented by Wei et al. (2026) included not detecting lesions when they were small, mistaking similar-looking diseases for one another, and misinterpreting regions due to shadows. The aforementioned shortcomings directly relate to this work. These show that fine-grained spatial segmentation remains a concern even for high-capacity teacher models on the PlantSeg disease dataset. These failure cases reinforce the need to analyze the compressed student not only on clean images but also under degraded input conditions (Wei et al., 2026). Selected synthetic corruptions provide a controlled way to examine whether clean-image segmentation improvements are retained under image degradation (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020). SegNeXt-B is chosen over ConvNeXt-L (Liu et al., 2022), which achieves a higher 46.24% mIoU on PlantSeg compared to 42.05% for SegNeXt-B, since the Multi-Scale Convolutional Attention Network (MSCAN) encoder (Guo et al., 2022) found in SegNeXt-B, unlike the ConvNeXt-L encoder with no hierarchy in terms of stages and spatial positions, produces a structured collection of hierarchical multi-scale feature maps, ideal for cascading the CNN-based SegNeXt backbone on top of the encoder stages with a channel-wise distillation approach, so as to extract and pass feature maps with improved scale information. While both SegNeXt-B and SegNeXt-L use an MSCAN encoder, they are differentiated by depth, whereby SegNeXt-B is more lightweight than SegNeXt-L (49M parameters). This work therefore leverages SegNeXt-B as a teacher, despite the slightly worse validation mIoU performance (44.52% for SegNeXt-L), given the similar number of channels to the student model and, importantly, no considerable increase in parameters beyond what is already gained in distillation through channel-wise interaction between the models (Wei et al., 2026).

MobileNetV3 and LR-ASPP

MobileNetV3, introduced by Howard et al. (2019), is a convolutional neural network architecture that is hardware-aware, meaning it was designed to take advantage of efficient CPU computation on mobile devices through depthwise separable convolutions, inverted residual blocks, and the NetAdapt algorithm for layer-by-layer efficiency optimization. One choice of key importance is Hard-Swish, which approximates the Swish nonlinearity while still being computationally fast (Howard et al., 2019). For the present study, MobileNetV3-Large is used with its native Hard-Swish and ReLU activations through the quantizable TorchVision implementation. Because that implementation already supports INT8 quantization of the native activation pattern, no global ReLU6 substitution is applied; replacing Hard-Swish would introduce an avoidable architecture change that may reduce FP32 accuracy without improving the selected INT8 workflow (Howard et al., 2019; Jacob et al., 2018; Krishnamoorthi, 2018).

Sensing studies of deep learning techniques for monitoring agricultural environment emphasize that even a lite version of a modern CNN such as MobileNetV3 with the Lite Reduced Atrous Spatial Pyramid Pooling (LR-ASPP) module (Howard et al., 2019) can match performance, using fewer FLOPs and parameters than those of models that depend on the full ASPP module (Howard et al., 2019). While river ice segmentation models were tested for comparable accuracy, the findings are relevant as evidence that LR-ASPP-style heads can support lightweight segmentation pipelines. Furthermore, the CPU-friendly design and hardware-specific neural architecture, coupled with an LR-ASPP head to support a high-resolution segmented output, are relevant to the present resource-constrained inference pipeline (Howard et al., 2019; Sola & Scott, 2022). In Sola and Scott (2022), although the LR-ASPP configuration showed performance trade-offs under degraded conditions when compared to related configurations, these findings support the inclusion of an evaluation section that quantifies performance loss under selected synthetic corruptions.

For this study, MobileNetV3-Large with its native Hard-Swish and ReLU activations (via the quantizable TorchVision implementation) is used for the student network. The MobileNetV3 result reported by Wei et al. (2026) is used only as contextual evidence for lightweight model difficulty on PlantSeg, not as the direct expected performance of the modified E1 baseline. The standard LR-ASPP head is task-adapted by expanding the low-level and high-level projections with an intermediate hidden layer before the final prediction head, while the final classifier is matched to the verified PlantSeg output label count (Chen et al., 2018; Howard et al., 2019). Exact parameter changes caused by this modified head will be computed from the implemented model and reported in the efficiency results. Furthermore, the encoder features generated at each hierarchical level of MobileNetV3's architecture are suitable for use with a Channel-Wise Knowledge Distillation strategy by using a training-only projection head to align selected student feature maps with teacher feature dimensions (Shu et al., 2021).

C. Knowledge Distillation for Model Compression

To address the notable computational demands of high-capacity segmentation architectures, researchers have increasingly explored Knowledge Distillation (KD) as a well-established approach to model compression. Hinton et al., (2015) popularized the concept of Knowledge Distillation, wherein a compact student network is trained to replicate the softened output probability distributions (soft labels) produced by a larger, well-trained teacher model rather than learning directly from hard ground-truth labels. This soft label transfer was shown to be effective in compressing knowledge from large, highly regularized models or model ensembles into significantly smaller deployable networks. However, Liu et al., (2019) identified that this logit-based transfer is insufficient for the dense prediction requirements of semantic segmentation, where accurate performance depends not only on categorical decisions but on the precise spatial relationships between each pixel in an image.

To address this, Liu et al., (2019) proposed Structured Knowledge Distillation (SKD), introducing two complementary distillation techniques: pair-wise distillation, which transfers pixel-to-pixel similarity relationships within feature maps, and holistic distillation, which captures the overall structural coherence of scene representations through adversarial training. Altogether, these mechanisms demonstrated that a compact student network could achieve segmentation performance comparable to architectures of considerably greater computational cost (Liu et al., 2019). As an alternative direction in dense-prediction distillation, Shu et al. (2021) proposed Channel-Wise Knowledge Distillation (CWD), which treats each channel of the teacher's and student's intermediate feature maps as a spatial probability distribution and minimizes the KL divergence between corresponding channel-wise distributions. Unlike pair-wise distillation, CWD does not require large spatial similarity matrices, and unlike holistic adversarial distillation, it requires no generator or discriminator network. These properties make CWD more computationally feasible for resource-constrained implementation settings while remaining specifically designed for dense prediction tasks such as semantic segmentation (Shu et al., 2021). Furthermore, a study by Hu et al. (2023) utilized Channel-Wise Knowledge Distillation for improving detection accuracy without increasing the number of parameters. Results showcase that their study was successful in enhancing the accuracy in helping detect Maize Leaf Disease, as their proposed algorithm which utilizes Channel-Wise Knowledge Distillation was able to reduce parameters by 15.5% showcasing the methods efficacy in both maintaining favorable accuracy while reducing parameter count. Another study by Saltık et al. (2025) utilizes Channel-Wise

Knowledge Distillation for improving lightweight weed detection. Results from the study demonstrate that the distilled student model was able to achieve a notable 2.5% improvement compared to other models. These studies provide evidence that Channel-Wise Knowledge Distillation can be effective for plant-disease imagery, supporting its consideration for the present study, which uses the PlantSeg dataset.

It should be noted that the agricultural applications cited above evaluate channel-wise distillation in object-detection and classification settings rather than in pixel-level segmentation (Hu et al., 2023; Saltik et al., 2025). These results therefore establish the plausibility of channel-wise distillation for plant-disease imagery and for the fine-grained, visually similar categories present in such data, but they do not by themselves demonstrate its effectiveness for dense lesion segmentation. Direct evidence for distillation applied to in-the-wild plant lesion segmentation remains limited, which is one of the gaps the present study is positioned to examine empirically rather than to assume. In addition, because the teacher and student belong to dissimilar architectural families—convolutional attention in SegNeXt versus an inverted-residual mobile CNN in MobileNetV3—channel-wise feature alignment is not guaranteed to transfer as cleanly as response-level guidance, since heterogeneous backbones tend to share comparable feature representations mainly at shallow stages. The E2-versus-E3 comparison is therefore treated not only as a measure of the marginal benefit of channel-wise distillation but also as a safeguard that would surface any negative transfer introduced by feature-level distillation; should such an effect appear, the Logit-KD-only configuration (E2) remains available as the distilled deliverable.

The present study does not adopt the full SKD framework of Liu et al. (2019). Pair-wise distillation is excluded because the $O(HW)^2$ spatial similarity matrices it requires are computationally prohibitive at 512×512 input resolution under the study's compute constraints (Liu et al., 2019). Holistic distillation is excluded because adversarial training introduces implementation complexity and training instability that are disproportionate to the scope of this proposed study (Liu et al., 2019). Instead, the proposed study will use Logit Knowledge Distillation (Hinton et al., 2015) in E2 as a response-level distillation baseline, and extends this with Channel-Wise Knowledge Distillation (Shu et al., 2021) in E3 to add feature-level channel guidance. This two-stage design allows the study to isolate the contribution of response-level teacher guidance (E2) from the additional benefit of channel-wise feature distillation (E3). Recent work cautions that comparisons among distillation methods can be confounded by training budget: when a well-tuned response-level (logit) baseline is trained long enough, much of the apparent advantage of more elaborate objectives can narrow, and some reported gains reflect differences in compute or schedule rather than the distillation signal itself (Beyer et al., 2022; Stanton et al., 2021; Zhao et al., 2022; Hao et al., 2023). For semantic segmentation specifically, a recent preprint argues that several segmentation-tailored distillation methods do not retain their advantage over canonical logit- and feature-based distillation once training compute is matched rather than iteration count; that analysis is demonstrated chiefly on memory-queue and cross-image relational methods and does not directly test channel-wise distillation (Ali et al., 2026). Channel-Wise Knowledge Distillation is comparatively lightweight: unlike pair-wise, holistic, or cross-image relational distillation, it adds only a channel-wise loss and a small training-only projection head, so its additional per-iteration cost over logit distillation is modest, and the present E2-versus-E3 comparison is designed to keep the training budget approximately fixed across the two stages (Shu et al., 2021). The need to report compute alongside accuracy, rather than relying on iteration count alone, is emphasized in efficiency-reporting guidance for deep learning (Dehghani et al., 2022). The teacher-student pairing uses SegNeXt-B / MSCAN-B as the practical high-capacity teacher, selected for its strong segmentation performance on standard benchmarks (Guo et al., 2022). The lightweight student is based on MobileNetV3 with a task-adapted Lite-ASPP-style head for efficient segmentation (Howard et al., 2019). Because an excessively large capacity gap between a high-capacity teacher and a lightweight student can itself limit the effectiveness of distillation, the present study treats teacher guidance as a means of narrowing rather than fully closing the student–teacher performance gap (Cho & Hariharan, 2019; Mirzadeh et al., 2020). In addition, because the teacher and student belong to dissimilar architectural families—the convolutional-attention design of SegNeXt versus the inverted-residual mobile CNN of MobileNetV3—channel-wise feature alignment is not guaranteed to transfer as cleanly as response-level guidance, as heterogeneous backbones tend to share comparable feature representations primarily at shallow stages. The comparison between E2 and E3 is therefore treated not only as a measure of the marginal benefit of channel-wise distillation but also as a safeguard that would surface any negative transfer introduced by feature-level distillation; should such an effect emerge, the Logit-KD-only configuration (E2) remains available as the distilled deliverable. Unlike Liu et al. (2019), who validated SKD on general scene segmentation benchmarks such as Cityscapes, the present study applies this distillation pipeline to the PlantSeg in-the-wild plant disease segmentation dataset, which presents additional challenges including extreme class imbalance, small lesion areas, and high inter-class visual similarity. The severe class imbalance and the predominance of small lesions in in-the-wild plant data also bear on the choice of training objective. Overlap-based losses such as the Dice loss are widely used for imbalanced pixel-level tasks because they normalize for region size and are less dominated by abundant background pixels than unweighted cross-entropy, and they are frequently combined with cross-entropy and with class-frequency weighting to stabilize learning under skewed label distributions (Müller et al., 2022; Isensee et al., 2021). This background motivates the compound, class-weighted supervised objective adopted in this study, the specific formulation of which is detailed in Chapter 3.

The selection of a MobileNetV3-based student in the present study's distillation pipeline is motivated by its well-documented structural suitability for resource-constrained visual recognition tasks. Originally proposed by Howard et al., (2019), MobileNetV3 achieves computational efficiency through depthwise separable convolutions and inverted residual blocks, enabling effective feature extraction at a substantially reduced parameter count relative to standard convolutional architectures. In the present study, the native Hard-Swish and ReLU activations of MobileNetV3 are retained throughout the backbone via the quantizable TorchVision implementation, which already supports INT8 quantization-aware training of these activations (Howard et al., 2019; Jacob et al., 2018). This efficiency-oriented design has since been validated across multiple independent plant disease recognition contexts (Zhang et al., 2025). In the present study, this decoder is task-adapted by expanding both its low-level and high-level classification projections with an intermediate 256-channel layer before the final output, following the intermediate projection convention established in segmentation decoder design (Chen et al., 2018), to accommodate PlantSeg's verified multiclass output space. Zhang et al. (2025) and Li et al. (2024) demonstrated that MobileNetV3 can be adapted for plant disease classification tasks, supporting its general suitability for plant disease image analysis. These studies do not directly evaluate MobileNetV3 on segmentation tasks, but they provide evidence that the backbone's core feature extraction is effective for plant disease imagery (Zhang et al., 2025; Li et al., 2024). Altogether, these studies support MobileNetV3 as a strong backbone and extensible foundation for plant disease feature extraction (Zhang et al., 2025; Li et al., 2024). The present study extends this capability to pixel-level segmentation through the integration of a Lite Reduced Atrous Spatial Pyramid Pooling (LR-ASPP) decoder (Howard et al., 2019). This positions the MobileNetV3-based student as a well-motivated lightweight architecture whose hierarchical feature structure is compatible with both Logit Knowledge Distillation and Channel-Wise Knowledge Distillation, as intermediate encoder feature maps can be extracted and aligned with the teacher's representations through a training-only projection head (Shu et al., 2021).

More recent distillation methods such as DIST (Huang et al., 2022), which uses Pearson correlation matching for stronger teacher-student gaps, were considered but not adopted as the primary method because their intra-class relational structure is expected to be less reliable under PlantSeg's extreme 115-class imbalance, where many disease categories appear in fewer than 20 training images (Wei et al., 2026). DIST is noted as a possible recovery method if Logit KD fails to produce improvement during E2 training.

D. Model Quantization for Resource-Constrained Inference: QAT and PTQ

Although the combined use of Logit Knowledge Distillation (Hinton et al., 2015) and Channel-Wise Knowledge Distillation (Shu et al., 2021) can effectively reduce the accuracy gap between a large teacher and a small student model, this method alone does not improve the fundamental precision of the network operations. Modern deep learning models primarily use 32-bit floating-point arithmetic (FP32). In the constrained memory and computation environment of resource-constrained hardware, executing FP32 calculations for high-resolution segmentation masks takes up more memory and CPU resources than lower-precision approaches, which has motivated the application of model quantization to the trained student model (Jacob et al., 2018; Howard et al., 2019). To lower the model's size and cost, models may be quantized by representing FP32 values with 8-bit integers (INT8) (Jacob et al., 2018).

A common approach to reducing model memory footprint is Post-Training Quantization (PTQ), which converts a pre-trained full-precision model to lower bit-width representations with minimal computational overhead and without requiring retraining. This approach allows for easy and fast quantization, but unlike QAT, the model is not trained with this effect exposed (Krishnamoorthi, 2018; Jacob et al., 2018). Therefore, PTQ can degrade accuracy more than QAT when it matters for precise localization, a key requirement for segmentation tasks and also for compression of models with architectures like MobileNetV3 with its many depthwise separable layers (Krishnamoorthi, 2018; Yang et al., 2021). Krishnamoorthi (2018) established that mapping full-precision weights and activations to lower bit-width without accounting for their underlying value distributions introduces quantization error that can meaningfully degrade model performance, a risk that becomes more consequential in tasks requiring fine-grained spatial discrimination (Krishnamoorthi, 2018; Nagel et al., 2021). In fact, one critical limitation for this type of compact architecture, which uses many depthwise separable convolutions like MobileNetV3, is that per-channel weight quantization should be preferred over per-tensor quantization of weights in its depthwise layers. This is because per-tensor weight quantization can cause high quantization error due to diverse weight distributions across individual depthwise channels (Krishnamoorthi, 2018). In this study, PTQ will be used as an E4/E7 baseline to justify the use of a 2×2 matrix of configurations to differentiate the effect of distillation on the PTQ versus QAT pipeline, as explained in the experimental design. For plant lesion segmentation, where boundary delineation directly determines the accuracy of disease severity estimation, this susceptibility to quantization-induced spatial degradation presents a practical concern (Ren et al., 2025).

To address the accuracy risks associated with PTQ, Jacob et al., (2018) established the foundational framework for Quantization-Aware Training (QAT). Rather than applying quantization as a post-processing step, QAT fundamentally modifies the training loop by introducing simulated quantization nodes (fake quantization nodes) that replicate the mathematical effects of INT8 precision during the forward pass. By exposing the network to this simulated precision loss while its weights are still being optimized, the model is able to gradually adapt its weight distributions to the constraints of reduced bit-width representations, mitigating the accuracy degradation that characterizes PTQ (Jacob et al., 2018). Jacob et al. (2018) demonstrated that models trained under this framework could be deployed using efficient integer-only arithmetic without dedicated floating-point hardware, closely preserving the accuracy of the original FP32 model. Krishnamoorthi (2018) further established that this training-time awareness of quantization constraints allows QAT to better preserve model accuracy relative to PTQ, particularly in tasks requiring fine-grained feature discrimination. This advantage has been examined empirically by Wu et al., (2020), who evaluated INT8 quantization across a range of deep learning inference tasks and reported that post-training quantization is often sufficient at INT8, while quantization-aware training more reliably recovers full-precision accuracy in the cases where post-training quantization is insufficient. While INT8 QAT shrinks the model and memory bandwidth and potentially speeds up inference depending on the backend, it has no impact on the theoretical FLOPs count, as the calculation logic is the same (Jacob et al., 2018).

Although the quantization framework reviewed above establishes the general viability of INT8 inference, much of the foundational evidence originates from image classification rather than dense prediction (Jacob et al., 2018; Krishnamoorthi, 2018; Wu et al., 2020). Direct evidence for low-precision semantic segmentation is comparatively limited. Nagel et al. (2019) noted that quantization results for segmentation architectures were largely unpublished at the time and reported that a DeepLabV3+ model with a MobileNetV2 backbone could be quantized to INT8 with less than a one-point reduction in mIoU on the Pascal VOC segmentation benchmark once distribution-aware techniques were applied. This finding is encouraging for the present pipeline, as it concerns an efficiency-oriented backbone of a similar family to the student used here; at the same time, it illustrates that segmentation-specific INT8 evidence rests on a narrower empirical base than the classification literature and is sensitive to quantization granularity, since per-tensor weight quantization of the same model degraded sharply before correction (Nagel et al., 2019; Krishnamoorthi, 2018). For this reason, the accuracy retained by the quantized student is treated in this study as a quantity to be measured empirically on PlantSeg rather than assumed from classification-domain results, and the difference between full-precision and quantized performance is reported as an empirical outcome rather than presented as lossless conversion.

Building on QAT research, in the present study, QAT is applied sequentially after the student model has gone through knowledge distillation (Hinton et al., 2015; Shu et al., 2021; Jacob et al., 2018). The QAT in this study is applied in E5 and E6. Standard QAT is applied to models that have been independently trained; in this study, QAT is instead employed to fine-tune the KD-compressed student model in E6 (Jacob et al., 2018). This design reflects the study's approach of first using knowledge distillation to transfer teacher guidance into the student model (Hinton et al., 2015; Shu et al., 2021). QAT is then used to adapt the student to simulated low-precision behavior (Jacob et al., 2018; Krishnamoorthi, 2018). PTQ is not used as a replacement for QAT but as a controlled experiment to establish a benchmark against QAT, that is, to evaluate how much better QAT can retain accuracy for the distilled model under resource constraints (Jacob et al., 2018; Krishnamoorthi, 2018).. This assessment is performed by evaluating the efficiency of the resulting models using CPU proxy benchmarks. Full INT8 inference will only be claimed after successful backend conversion and operator-support verification (Jacob et al., 2018; PyTorch, n.d.). If some operators of the student model or the adapted task-specific decoder head remain in floating point after conversion, these configurations are presented as partially quantized rather than fully INT8.

E. Evaluating Model Robustness in the Wild

Beyond the accuracy and efficiency considerations of the compression pipeline established above, a third evaluation dimension concerns model behavior under degraded input conditions. A recognized challenge in the deployment of deep learning models is their susceptibility to performance degradation under real-world distributional shifts which are conditions that differ meaningfully from the clean, controlled datasets on which models are typically evaluated (Liu & Jin, 2023). While a model may achieve strong Mean Intersection over Union (mIoU) performance in a laboratory setting, this metric alone often fails to reflect its reliability under actual deployment conditions (Kamann & Rother, 2020). Hendrycks and Dietterich (2019) formalized this concern in the context of image classification, demonstrating through the ImageNet-C benchmark that neural networks exhibit significant performance degradation when subjected to common real-world image corruptions (e.g motion blur, digital noise, weather-related artifacts), establishing a foundational framework for evaluating model robustness under controlled corruption severities. Building from this foundation, Kamann and Rother (2020) extended robustness benchmarking specifically to semantic segmentation, establishing that dense prediction tasks are uniquely susceptible to environmental perturbations due to their dependence on precise pixel-level feature discrimination. To quantify this vulnerability systematically, they formalized the use of Corrupted mIoU (mIoU-C) as a metric for measuring segmentation accuracy across varying severities of image distortion a methodology that the present study adopts to evaluate the robustness of the proposed compression pipeline under selected synthetic corruption conditions applied at test time. In addition to mIoU-C, the present study reports the Relative Performance Drop (RPD), defined as the percentage decrease in mIoU between clean and corrupted test conditions, to quantify how much segmentation performance is lost under the selected corruptions. A lower RPD indicates better performance retention. RPD is reported as a descriptive measure and does not constitute proof of complete real-world robustness (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020). As a complementary descriptive measure, relative corruption degradation (rCD) may also be reported to express each model's corruption-induced degradation relative to an internal reference model, isolating robustness differences from clean-accuracy differences; because this reference is internal to the study rather than an external fixed architecture, rCD is interpreted as a within-study relative measure rather than a cross-study comparison (Kamann & Rother, 2020).

Within the specific domain of agricultural computer vision, this issue of dataset bias has historically limited the real-world utility of plant disease models (Ferentinos, 2018). Early agricultural datasets predominantly featured single leaves photographed against uniform, contrasting backgrounds under ideal lighting conditions (Mohanty et al., 2016). To address this significant limitation, Wei et al. (2026) introduced PlantSeg, a large-scale, "in-the-wild" dataset explicitly designed for plant disease segmentation. PlantSeg provides pixel-level ground-truth annotations for images captured in complex, unconstrained environments, featuring overlapping foliage, variable illumination, and diverse background noise.

This study applies this robustness assessment framework to assess the proposed compression pipeline. First, it utilizes the PlantSeg dataset so that baseline models are trained on complex, in-the-wild representations rather than sterile, laboratory-grade images (Wei et al., 2026). Furthermore, this study integrates corruption-based robustness benchmarking principles into its testing phase (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020).

Following Hendrycks and Dietterich (2019), this study selects five representative image corruptions: Gaussian noise, JPEG compression, motion blur, brightness, and fog. The five corruptions are selected to approximate distinct and commonly encountered image-acquisition disturbances rather than to enumerate every condition a deployed system might face. Motion blur and Gaussian noise correspond to handheld capture and to low-light or low-cost sensors; JPEG compression reflects the lossy encoding applied when images are stored or transmitted from the field; brightness variation represents uncontrolled outdoor illumination; and fog-like degradation approximates haze or condensation along the optical path (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020). These correspondences are offered to motivate the selection only and are not claims about the prevalence of any condition in the field. These corruptions are added to the PlantSeg test set at severity levels 1–3. They are applied to the raw pixels of test images before normalization, while ground-truth segmentation masks are not altered. The robustness evaluation is applied to representative student model configurations to isolate the impact of knowledge distillation and quantization on performance under degraded image inputs. These corruptions are selected as controlled synthetic test-time conditions and are not treated as a full representation of all real agricultural field conditions; robustness to synthetic corruptions has been shown not to transfer reliably to natural distribution shifts, so these results are interpreted as a controlled stress test rather than evidence of field readiness (Taori et al., 2020).

Evaluating the robustness of the compressed pipeline specifically, rather than the full-precision baseline alone, is warranted because model compression can itself alter behavior under corrupted inputs. In image classification, quantized networks have been reported to be more susceptible to common corruptions than their full-precision counterparts, indicating that low-bit representations do not automatically preserve robustness (Xiao et al., 2023). For semantic segmentation, knowledge distillation has been shown to improve corrupted-input performance when transferring representations from a high-capacity teacher to a compact student (Amirkhani et al., 2021). These results originate from a classification-domain quantization study and from a distillation method designed to improve robustness, respectively, rather than from a neutral benchmark of combined quantization and distillation under corruptions for plant lesion segmentation; the interaction between INT8 quantization and corruption robustness in dense prediction therefore remains under-characterized and is examined empirically in the present study rather than assumed.

To accurately quantify a model's robustness to real-world perturbations, the present study prioritizes evaluation metrics explicitly designed for dense prediction tasks. While standard classification accuracy is sufficient for binary image-level diagnosis, it is fundamentally inadequate for assessing spatial boundary quality (Kamann & Rother, 2020). To address this, the present study adopts Mean Intersection over Union (mIoU) as its primary segmentation evaluation metric. Müller et al., (2022) established mIoU as a well-adopted standard in medical image segmentation -- a domain that shares with agricultural diagnostics the fundamental requirement for pixel-precise delineation of pathological regions, where boundary accuracy is closely tied to diagnostic reliability. This structural similarity in task requirements supports the applicability of mIoU as an equally appropriate evaluation standard in agricultural lesion segmentation contexts, where mIoU is the standard segmentation metric reported on plant disease benchmarks (Wei et al., 2026). The metric rigorously penalizes both false positives, predicting disease where healthy tissue exists, and false negatives, failing to identify actual necrotic areas, making it sensitive to the spatial errors most consequential for disease severity estimation (Müller et al., 2022; Ren et al., 2025). In the case of datasets recorded under real conditions, such as PlantSeg, assessing compressed models beyond clean-image accuracy is appropriate (Wei et al., 2026). This assessment includes robustness to common image-acquisition problems represented by the selected synthetic corruptions. (Hendrycks & Dietterich, 2019; Kamann & Rother, 2020).

Alongside mIoU, the Dice coefficient is reported as a secondary overlap metric. Dice and IoU are monotonically related at the class level, so the two measures generally yield consistent model rankings; Dice is nonetheless retained for two reasons. It forms part of the compound supervised objective used to train the student models, and its region-overlap normalization is informative for the small and localized lesions that characterize PlantSeg, in which the majority of images contain comparatively small lesion masks (Müller et al., 2022; Wei et al., 2026; Ren et al., 2025). Because of its close relationship with IoU, Dice is interpreted as a complementary descriptive measure rather than as an independent indicator of segmentation quality.

F. Synthesis

The literature reviewed in this chapter establishes the foundation for the present study as it exposes both the trajectory of digital agriculture and the unresolved bottlenecks that limit its real-world application. Each reviewed concept contributes directly to the proposed pipeline and justifies the study's focus on the accuracy–efficiency–robustness trade-off.

Table 2.2 Synthesis of Related Studies and Architectural Proof

Author & Year	Architecture	Subject Focus	Performance Metric
Guo et al. (2022)	SegNeXt	High-capacity Semantic Segmentation (Teacher model foundation)	mIoU / Spatial Accuracy

Howard et al. (2019)	MobileNetV3 + LR-ASPP	Primary basis for the lightweight segmentation student architecture	Parameters / mIoU
Zhang et al. (2025)	MobileNetV3-based plant disease model	Supporting evidence that MobileNetV3 can be adapted for plant disease image analysis, but not direct evidence for PlantSeg segmentation	Accuracy / Parameter Count
Hinton et al. (2015)	Logit Knowledge Distillation	Response-level teacher guidance for student training (E2 baseline)	mIoU / Accuracy
Shu et al. (2021)	Channel-Wise Knowledge Distillation (CWD)	Channel-wise feature distribution alignment for dense prediction distillation (E3)	mIoU
Jacob et al. (2018)	Quantization-Aware Training (QAT)	INT8 Quantization-Aware Training to reduce model size and preserve accuracy under lower-precision inference (E5, E6)	Inference Latency / Accuracy
Wei et al. (2026)	PlantSeg Dataset	Providing complex, unconstrained "in-the-wild" agricultural data	mIoU / mAcc (benchmark results used as contextual evidence only)
Kamann & Rother (2020)	Robustness Benchmarking	Evaluating dense prediction models against environmental perturbations	Corrupted mIoU (mIoU-C)
Hendrycks & Dietterich (2019)	Corruption Benchmark	Common image corruption taxonomy for robustness evaluation	Corrupted accuracy / mIoU

First, the reviewed literature on agricultural diagnostics clearly defines the core problem: early classification models laid the groundwork for automated disease detection, but they lack the spatial intelligence required for modern agricultural intervention (Wang et al., 2025; Das et al., 2024). Semantic segmentation offers an effective approach for isolating the plant lesions (Ren et al., 2025). It can be confirmed that high-capacity segmentation architectures, such as SegNeXt, use too many resources to serve as the resource-constrained student model in this study (Guo et al., 2022). Hence, teacher-student distillation is used to transfer guidance from the high-capacity model to the lightweight student (Hinton et al., 2015; Shu et al., 2021). INT8 quantization is then used to support lower-precision model compression under resource-constrained conditions (Jacob et al., 2018; Krishnamoorthi, 2018).

Logit Knowledge Distillation offers a solid choice for knowledge transfer from teacher to student and serves as a good basis for comparison (Hinton et al., 2015). To achieve performance gains specifically for dense prediction tasks, such as semantic segmentation, it is additionally helpful to align the channel-wise representations between the teacher and the student network, as presented by Channel-Wise Knowledge Distillation (Shu et al., 2021) . These methods constitute stages E2 and E3 in the ablation experiments. However, model compression through architecture alone does not address numerical precision. With that, Quantization-Aware Training (Jacob et al., 2018) is included as the primary approach to solve this problem, stages E5 and E6, while Post-Training Quantization (PTQ) is retained as an empirical baseline, stages E4 and E7, to compare adaptive quantization during training with post-training calibration in preserving segmentation accuracy (Krishnamoorthi, 2018).

Moreover, based on the empirical observations by Hendrycks and Dietterich (2019) and Kamann and Rother (2020) that models trained solely on clean datasets underperform under diverse corruption scenarios, this study uses in-the-wild data from the PlantSeg dataset (Wei et al., 2026). Therefore, to make a comparative study on corruption resilience, the test set of the PlantSeg dataset is supplemented with common synthetic corruptions in severity levels 1 to 3 (Hendrycks & Dietterich, 2019). This standard evaluation process uses established performance indicators such as corrupted mIoU (mIoU-C) and the Relative Performance Drop (RPD) that reflect the degree to which the models maintain their segmentation quality when faced with inputs affected by noise, JPEG compression, blur, varying brightness, and fog.

Since the PlantSeg dataset is a relatively recent benchmark, its use for compressed plant lesion segmentation remains underexplored (Wei et al., 2026). The specific combination of Logit Knowledge Distillation, Channel-Wise Knowledge Distillation, and INT8 Quantization-Aware Training is grounded in established compression methods (Hinton et al., 2015; Shu et al., 2021). INT8 Quantization-Aware Training is grounded in low-precision quantization research (Jacob et al., 2018). The present study is intended as an initial, controlled investigation into these established compression methods for PlantSeg, rather than as a novel theoretical contribution. Instead, it adopts an integrated approach by leveraging established distillation and quantization techniques to investigate resource-constrained compression of a lightweight student model (Hinton et al., 2015; Jacob et al., 2018). The resulting trade-off is evaluated in terms of accuracy, efficiency, and robustness within the PlantSeg benchmark context (Wei et al., 2026; Kamann & Rother, 2020)